

Quiz (Habanero)

- Q1.** Solve the inequality $2x - 5 > 11$.
(A) $x > 8$
(B) $x < 8$
(C) $x > 4$
(D) $x < 4$
(E) $x = 8$
- Q2.** Graph the solution to $x + 2 \leq 7$ on a number line.
(A) $(-\infty, 5]$
(B) $[-\infty, 5]$
(C) $[5, \infty)$
(D) $(-\infty, 5)$
(E) $[5, \infty]$
- Q3.** Write the solution to $x - 3 < 9$ in interval notation.
(A) $(-\infty, 12)$
(B) $(-\infty, 6)$
(C) $[6, \infty)$
(D) $(-\infty, 6]$
(E) $[6, \infty]$
- Q4.** If the temperature is 15°C and rises by 2°C every hour, when will it reach 25°C ?
(A) 5 hours
(B) 6 hours
(C) 4 hours
(D) 3 hours
(E) 2 hours
- Q5.** Solve the inequality $4x + 2 > 3x + 5$.
(A) $x > 3$
(B) $x < 3$
(C) $x > 1$
(D) $x < 1$
(E) $x = 3$
- Q6.** If the rainfall in a forest is x mm and the maximum allowed is 200 mm, write an inequality to represent x .
(A) $x \leq 200$
(B) $x \geq 200$
(C) $x < 200$
(D) $x > 200$
(E) $x = 200$
- Q7.** Graph the solution to $2x - 1 \geq 5$ on a number line.
(A) $[3, \infty)$
(B) $(-\infty, 3)$
(C) $(-\infty, 3]$
(D) $[3, \infty]$
(E) $(-\infty, 3)$
- Q8.** If the temperature drops by 5°C every hour and is currently 20°C , when will it reach 0°C ?
(A) 5 hours
(B) 4 hours
(C) 3 hours
(D) 2 hours
(E) 1 hour

Open Ended Questions (FRQ)

- Q9.** A certain ecosystem can thrive when the temperature is between 10°C and 25°C . Write an inequality to represent this situation and graph the solution on a number line.
- Q10.** The amount of rainfall x mm in a given month must be greater than 50 mm for a certain type of plant to grow. Write this situation as an inequality and solve for x . Provide your answer in interval notation.

Chef's Tips

- Provide clear graphs for the free response questions.
- Use the number line to check the solutions for the multiple choice questions.
- Make sure to label the axes clearly when graphing the solutions.